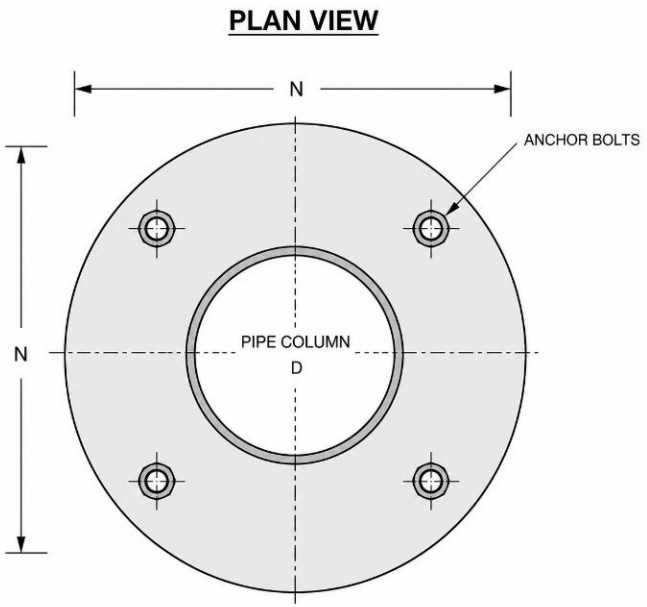
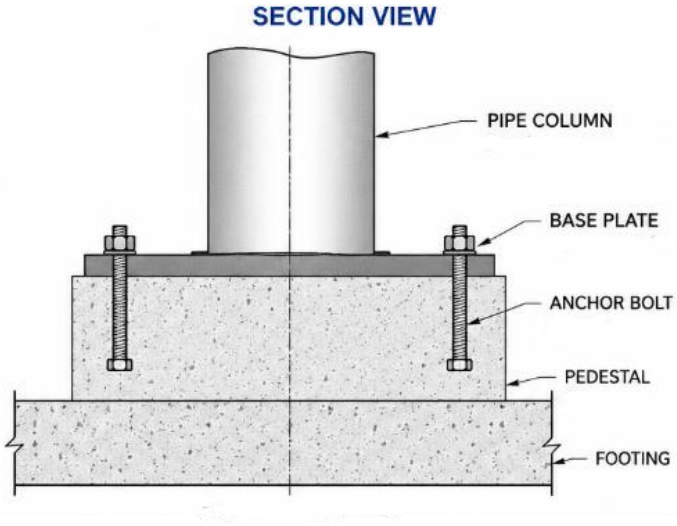
2. Base plate segment

Design Parameter	Symbol	Value	Source
Factored axial load (LRFD)	P_u	4623.218 kips	Main Tank Design (Predefined)
Governing overturning moment (LRFD)	M_u	3823.08 kip-ft	Main Tank Design (Predefined)
Pipe diameter	D_{pip}	62.4134 in	Main Tank Design (Predefined)
Number of legs	N_{legs}	6	User Input
Number of anchor bolts	N_b	4	From Anchor Bolt Design
Anchor bolt diameter	d_b	1.5 in	From Anchor Bolt Design
Anchor bolt hole diameter	d_b	2 in	From Anchor Bolt Design
Bolt arrangement dimensions	b×l	38in × 12 in	From Anchor Bolt Design
Base-Plate cross-section	D_{co}	34	
Pedestal cross-section	P×L	54 in × 38 in	From Anchor Bolt Design
Concrete compressive strength	f'_c	4 ksi	User Input
Steel yield strength	F_y	36 ksi	User Input
Modulus of elasticity	E	29,000 ksi	AISC Material Property
LRFD resistance factor (plate bending)	ϕ_b	0.90	AISC LRFD
Pipe Column Diameter	Dpipe	24”	From Main Tank (Predefined)





1. Determine the Factored Moment.

$$\mathrm{M_{u,ped}} = \frac{M_u}{\mathrm{N_{ped}}}$$

$$\mathrm{M_{u,ped}} = \frac{3823.08}{6}$$

$$\mathrm{M_{u,ped}} = 637.18 \text{ kip-ft}$$

Convert to kip-in:

$$\mathrm{M_{u,ped}} = 637.18(12)$$

$$\mathbf{M_{u,ped} = 7646.16 \text{ kip-in}}$$

2. Pedestal Area

$$\mathrm{A_2} = (54)(38)$$

$$\mathrm{A_2} = 2052 \text{ in}^2$$

Base Plate Area

$$\mathrm{A_1} = \frac{\pi D^2}{4} = \mathrm{A_1} = \frac{\pi(34)^2}{4}$$

$$\mathrm{A_1} = 907.9203\ in^2$$

Compute Bearing Stress

$$\mathrm{F_p} = 0.85\phi_c \mathrm{f'_c} \sqrt{\frac{\mathrm{A_2}}{\mathrm{A_1}}}$$

$$F_p = 0.85(0.65)(4)\sqrt{\frac{2052}{907.9203}}$$

$$F_p = 3.322 \text{ ksi}$$

Check Limit

$$1.7\phi_c f'_c = 1.7(0.65)(4) = 4.42 \text{ ksi}$$

$$3.24 < 4.42 \ldots\ldots\ldots\ldots\ldots\ldots \textbf{PASS}$$

3. Assume Base Plate Cross-Section

Adopt:

$$N = B = 34 \text{ in}$$

Base plate area:

$$A_1 = \frac{\pi(34)^2}{4}$$

$$A_1 = 907.9203 \text{ in}^2$$

Check Bearing Capacity

$$P_n = F_p A_1$$

$$P_n = (3.322)(907.9203)$$

$$P_n = 3016.510 \text{ kips}$$

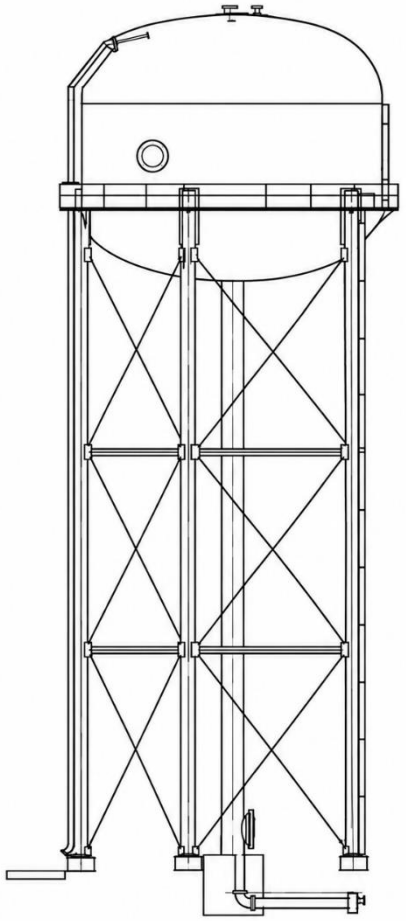
Applied load:

$$P_u = 1541.073 \text{ kips}$$

Note: for this scenario we considered half of the pedestals in compression and half of them in tension as show in the photo

$$3016.510 > 1541.073$$

PASS



4. Determine Equivalent Eccentricity

$$e = \frac{M}{P}$$

Substitute:

$$e = \frac{7646.160}{770.536}$$

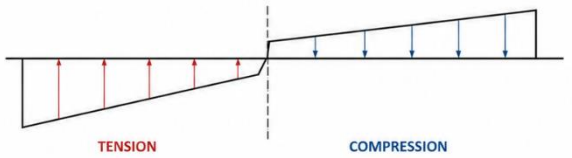
$$e = 9.923 \text{ in}$$

Check Bearing Condition

For full compression:

$$e \leq \frac{N}{6}$$

$\frac{N}{6} = \frac{34}{6} = 5.667$ in……. if this cause controls Pu/(1.0*N)



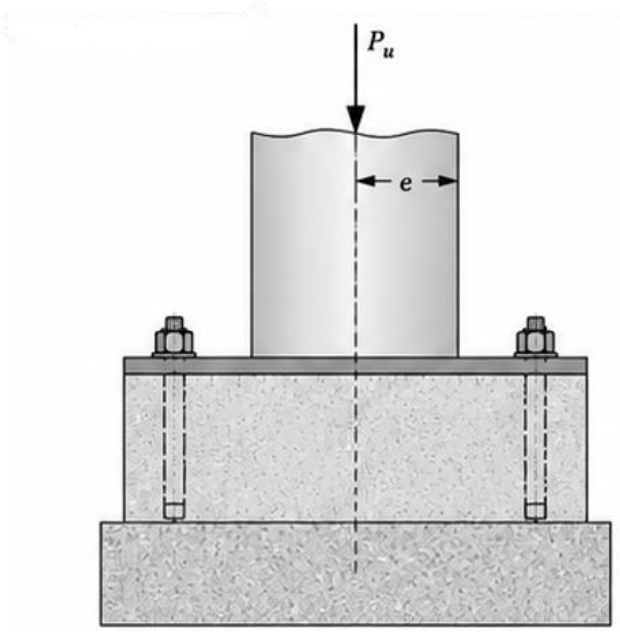
Since:

$9.923 > 5.67$……..if this cause controls (Pu/0.5*N)

Full compression does not occur.

Therefore:

- Uplift develops.
- Anchor bolts resist tension.
- Partial compression exists beneath the base plate.



$$\boxed{e = 9.923 \text{ in} > \frac{34}{6} = 5.667 \text{ in}}$$

5. Determine Resultant Anchor Bolt Force

From the completed anchor bolt design:

$$T_{leg} = 103.667 \text{ kips}$$

Number of anchor bolts:

$$N_b = 6$$

Anchor force per bolt:

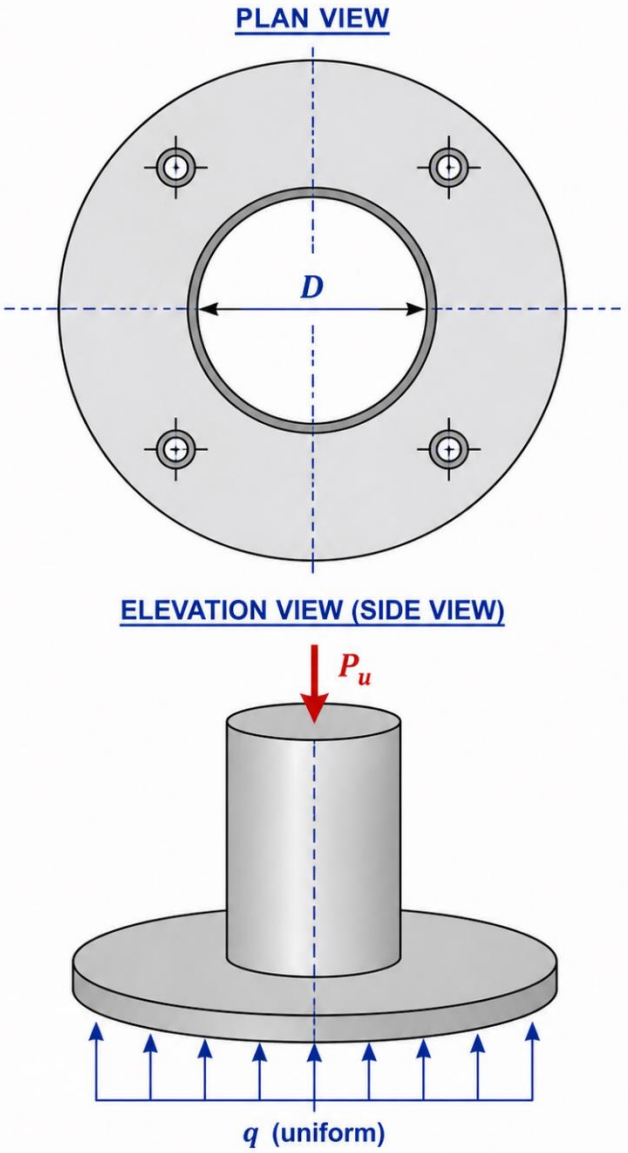
$$T_{bolt} = \frac{T_{leg}}{N_b} = 25.916$$

6. Determine Base Plate Thickness

6.1. Determine Bearing Area of Pipe Column

Cylinder diameter:

$$D = 24 \text{ in}$$



Column bearing area:

$$A_b = \frac{\pi D^2}{4}$$

$$A_b = \frac{\pi(24)^2}{4}$$

$$A_b = 452.389 \text{ in}^2$$

6.2. Determine Bearing Pressure

$$q = \frac{P_u}{A_b}$$

$$q = \frac{770.536}{452.389}$$

$$q = 1.703 \text{ ksi}$$

$$\mathbf{q = 1.703 \text{ ksi}}$$

6.3. Determine Plate Projection

$$m = \frac{N - D}{2}$$

$$m = \frac{34 - 24}{2}$$

$$\mathbf{m = 5\ in}$$

6.4. Determine Plastic Moment per 1-in Strip

$$M_{plu} = \frac{qm^2}{2}$$

$$M_{plu} = \frac{(1.703)(5)^2}{2}$$

$$M_{plu} = 21.291 \text{ kip-in/in}$$

6.5. Determine Required Thickness

Using AISC LRFD:

$$t_p = \sqrt{\frac{4M_{plu}}{0.90F_y}}$$

$$t_p = \sqrt{\frac{4(21.291)}{0.90(36)}}$$

$$t_p = \sqrt{\frac{85.163}{32.40}}$$

$$t_p = \sqrt{2.6285}$$

$$t_p = 1.621 \text{ in}$$

Adopt: $\underline{\mathbf{t_p = 1.70\ in}}$